Advanced Database Systems, Projects 2026

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1 General terms

1. Groups. There will be 2-person groups.
2. Topic. Each group works on the same topic independently.
3. Working environment. Each group will get a separate GitLab project, which should be used to manage the project (board/issues if you need them), document it and store the code and other data. All relevant files should be in the main branch. Before getting the project repository you **must** go to `https://gitlab.kis.agh.edu.pl` and login with SSO AGH.
4. Choice of technology. Unless stated otherwise, the chosen technologies and architecture should be as light as possible, e.g. require minimum

setup, be easy to launch and evaluate. Projects must run under Linux. Architectural choices have to be approved by the instructor. Preferably free software or open-source. Use of any more restrictive license must be confirmed with the instructor.

5. Self-sufficiency. Projects should be self-sufficient, e.g. include code needed to set up the dependencies, create database structures and populate them with data. Source data should be collected from external sources by a script or included in the repo.

6. Documentation. Projects must include all documentation in `README.md` or `README.org` file in the root directory of branch main in the form of Markdown or org-mode files respectively. It should contain instructions on how to deploy and run the software, define the prerequisites and document the project design and implementation process. It has to state the roles of all the students in the project and description of who did what. The requirements for the documentation are listed in the Doc Checklist section.

7. Results. The results should be easy to evaluate, and reproducible. A step-by-step manual on how to reproduce the results has to be provided.

8. Self-evaluation. Self-evaluation of the projects is mandatory. Efficiency should be discussed, along with proposed strategies for future mitigation of identified shortcomings.

2 Grading policy

1. The projects will be graded based on the items 4-8 of the General Terms and technical merit.

2. Each group has to consult the project progress in class.

3. The project outcomes must be presented during class as the final presentation.

4. You can make up the project twice after the semester ends. The makeup schedule will be announced at the end of the semester. All makeups are graded and the grades remain on your record.

grade	%
5	90
4.5	80
4	70
3.5	60
3	50
2	

3 Docs Checklist

Make sure that the documentation consists of the items below.

1. ☐ Choice of technology.
2. ☐ Architecture: components and interactions, optional diagram.
3. ☐ Prerequisites (software modules, databases, etc.).
4. ☐ Installation and setup instructions.
5. ☐ Design and implementation process, step by step.
6. ☐ Details on how each of the goals is addressed, including database queries and logic behind them.
7. ☐ The roles of all the students in the project and description of who did what.
8. ☐ Results, including example runs, outcomes and timings indicating efficiency.
9. ☐ User manual (how to run the software) and a step-by-step manual how to reproduce the results.
10. ☐ Self-evaluation: efficiency should be discussed, strategies for future mitigation of identified shortcomings.

4 Topic and Datasets

Starting points:

- general description: `https://en.wikipedia.org/wiki/Vector_database`
- datasets: `https://github.com/vector-index-bench/vibe`

5 Goals/Queries

1. ML meaningful queries regarding neighborhood, distance and smilarity (DQL), e.g.:
   - KNN,
   - K-Means.
2. All goals must be numbered and their parameters must be specified.
3. A single CLI app that runs particular query: `db`
   - e.g. running query #1 with two parameters: `db 1 443321 4`,
   - number of paramters can vary from query to query.

   Details TBD.

6 Measuring performance

All groups must determine how to assess the resulting implementation in a common way to have comparable performance results among groups. It will be discussed in class. In particular:

- what quantities should be measured,
- which queries to use,
- what specific values of query parameters to use,
- do we measure anything else apart from the queries?

Details TBD.

7 Schedule

1. Intro, teams, repos.
2. Datasets, technologies and deployment (mind that relationship with the performance measures: how to deploy, how to compare results, docker/podman).
3. Database design, populating, tools (DDL, tooling).

4. Queries/goals (DQL, tooling).
5. Performance measures (based on #4, isolated server?).
6. Problems, tweaks, preliminary outcomes.
7. Final Presentations (software and docs must be uploaded to the repo before the presentation).